

From input to inspectable evidence

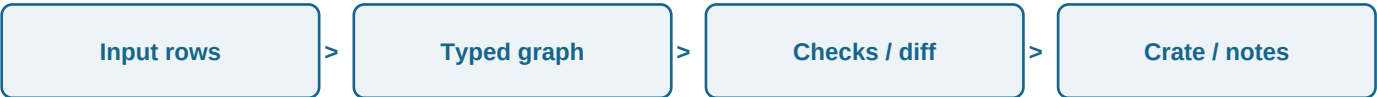
A practical workbook for ExampleCo-GHG-001

GHG Assurance Graph | 0.4.0a1 source revision | 29 September 2026

2025-v1		3820
2025-v2		3850
2026-v1		4060

Three selected-source inventories, 45 calculations and 73 recorded CLI actions. Figures are derived from executed JSON outputs, not a simulated application interface.

Read sequentially: prepare (2-3), inspect (4-7), investigate (8-10), preserve and share (11-13), experiment (14), close (15-16).



INTERPRET / All organizations, source documents, factors and review labels are invented. This is not a complete inventory or human assurance.

NEXT / Prepare a local locked Python environment.

01 / Prepare a locked environment

Run these commands in your terminal.

Prerequisites: Git, uv and Python 3.12. The package requires $\geq 3.12, < 3.13$; this run used 3.12.14. Installation may download dependencies. No API key or paid service is required.

```
git clone https://github.com/simon5530/ghg-assurance-graph.git
cd ghg-assurance-graph
git checkout main
uv sync --locked
git rev-parse HEAD
```

Use the source commit accompanying this guide, not an older release tag. Record the printed SHA for exact reproduction. uv.lock pins dependencies; do not substitute an unpinned installation.

Check	Expected
Python	3.12.x
Environment	.venv created by uv sync
Working directory	Repository root
Run network	Process-local socket guard

INTERPRET / The run is offline after installation. The guard is not an OS sandbox; installed dependencies are trusted.

NEXT / Choose a new or empty output folder.

02 / Execute the complete workflow

One command creates the evidence tree.

```
uv run --locked python scripts/run_example.py
  --out artifacts/worked-example
```

The two lines above are one command: join with a space. Open artifacts/worked-example/REPORT.md first. In this guide OUTPUT means that output folder.

Stage	Saved evidence
Generate / calculate	generated/ and calculations/
Trace / validate	stages/<snapshot>/
Compare / declare	diffs/ and declarations/
Package / verify	packages/ and package-verify.json
Export / inspect	vaults/ and REPORT.md

```
{
  "commands": 73,
  "source_records": 45,
  "totals_kg_co2e": {
    "2025-v1": 3820,
    "2025-v2": 3850,
    "2026-v1": 4060
  }
}
```

INTERPRET / Actual summary.json excerpt. Success checks 45 Fraction products, literal totals, expected failures, package integrity and vault links.

NEXT / If the destination exists, choose a new name instead of deleting evidence.

03 / Map the input fields

Open OUTPUT/generated/2025-v1/inputs.json.

Field / role	Actual electricity input
id / matching key	electricity
activity + unit	1000 kWh
factor + denominator	0.5 kgCO2e / kWh
allocation share	1
scope / basis	2 / location-based
year / geography	2025 / TW
factor basis	precharacterized-kg-CO2e
apply_gwp	false
origin / review	primary / synthetic-accepted

Also inspect factor_source, factor_year, evidence, method, gwp_basis, uncertainty and coverage. reported_kg_co2e is an independently checked result, not an extra operand.

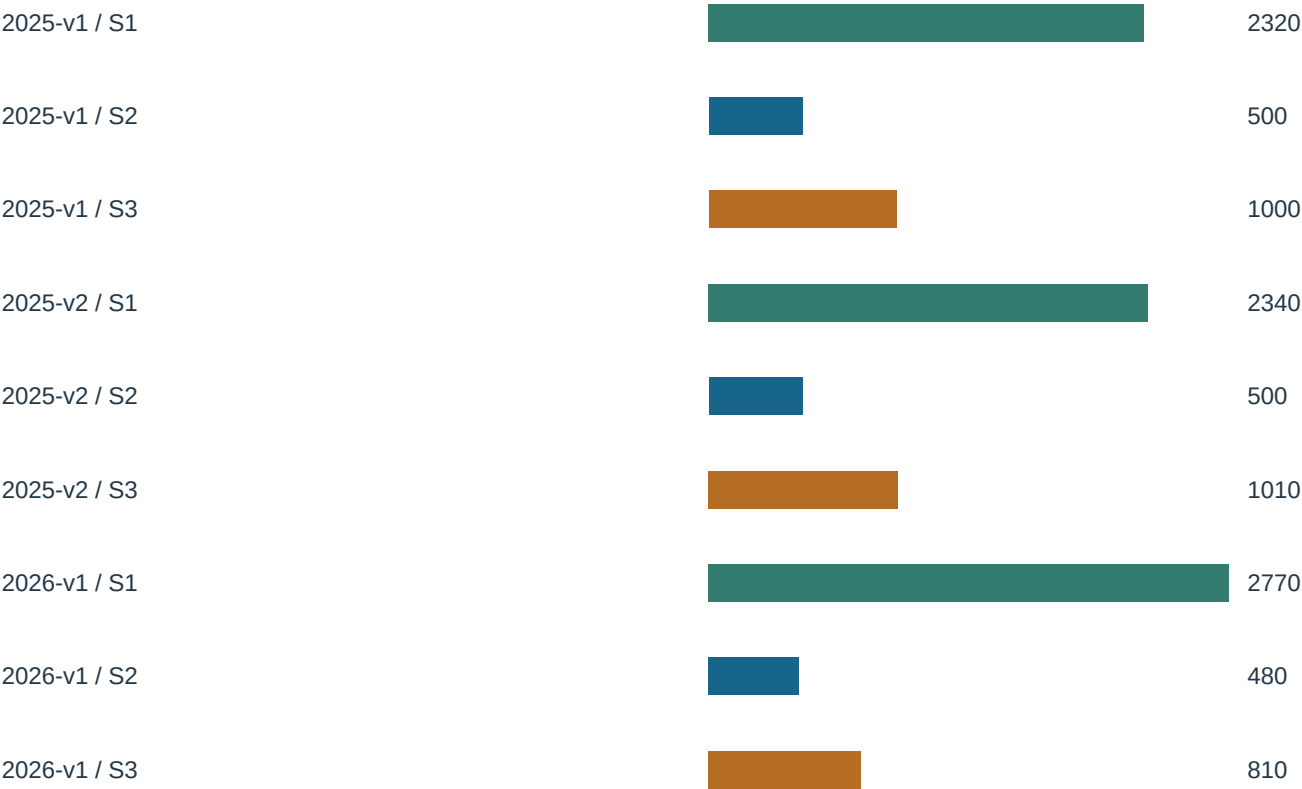
INTERPRET / $1000 \text{ kWh} \times 0.5 \text{ kgCO}_2\text{e/kWh} \times 1 = 500 \text{ kgCO}_2\text{e}$. Precharacterized factors already include GWP: do not multiply by GWP again.

NEXT / Compare snapshots without calling their difference verified abatement.

04 / Read totals and composition

Compare actual selected-source results, in kgCO2e.

Snapshot	Scope 1	S2 LB	Scope 3	Total
2025-v1	2320	500	1000	3820
2025-v2	2340	500	1010	3850
2026-v1	2770	480	810	4060



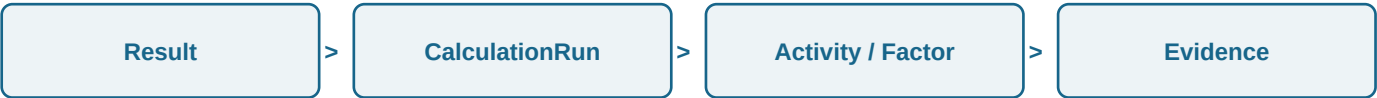
Labels, not color alone, identify scopes. Scope 2 is location-based only; market-based results are alternatives, not additive totals.

INTERPRET / 2025-v2 restates 2025-v1 (+30). 2026-v1 is a changed-basis scenario (+210), not a harmonized performance claim. Scope 3 is partial.

NEXT / Trace a 500 kgCO2e result before trusting an aggregate.

05 / Follow a typed lineage

Open OUTPUT/stages/2025-v1/explain/electricity.json.



Typed node	Local ID suffix / role
EmissionResult	electricity-result:v1 / 500 kgCO2e
CalculationRun	electricity-run:v1
ActivityRecord	electricity-activity:v1 / 1000 kWh
EmissionFactor	electricity-factor:v1 / 0.5
CalculationMethod	electricity-method:v1
GWPSet	electricity-gwp:v1
EvidenceArtifact	electricity-evidence:v1
ReviewDecision	electricity-review:v1 / accepted

Reference direction: result.calculation -> run; run.activity -> activity; run.factor -> factor. Both operands reference synthetic evidence. Run also references method, GWP, boundary and inventory.

```
Shared ID prefix:
urn:ghgag:exampleco-ghg-001-2025-v1-
Actual explain value: 500.0 kg_CO2e
```

The same row JSON supports activity and factor evidence here: reproducible provenance, not independent corroboration.

INTERPRET / Accepted is a simulated status, not authenticated reviewer authority. Snapshot IDs are not cross-year revision edges.

NEXT / Inspect clean and failing validation reports.

06 / Validate, including failures

Do not interpret exit 0 as assurance.

```
{
  "conforms": true,
  "findings": []
}
```

Control / expected exit	Finding and target
Clean / 0	conforms=true; no selected findings
Double GWP / 1	refrigerant: no double GWP
Missing source / 1	natural-gas: factor provenance

Double GWP: for a precharacterized factor set `apply_gwp=false` on the source row, then regenerate and revalidate. Do not delete the finding.

Missing source: restore a legitimate `factor_source` citation. In this fixture, regenerate the invented source record. In real use, stop until evidence is available.

```
Full finding IDs:
urn:ghgag:finding-f9726656c060f3f707e6cf1de9e8f56e0c9c9407242638a0f561226fc46c678b:v1
urn:ghgag:finding-c56b16b595723227a3409ec693fd0b76d8cbce7cc1bc26df91de979b688e5c36:v1
```

INTERPRET / Two exit-1 controls are expected. Ten defects exist; the full tests evaluate all. Clean means only that encoded checks passed.

NEXT / Keep failure receipts and separate restatement from annual change.

07 / Read a same-year restatement

Open both 2025-v1--2025-v2 diff files.



Mode	Delta	Signed UNKNOWN	Absolute UNKNOWN
Unassisted	30.0	10.0	10.0
Assisted	30.0	0	0

Fleet: 30 -> 40 liters at factor 2 gives +20. Travel: 100 -> 150 km at factor 0.2 gives +10. Total 3820 -> 3850.

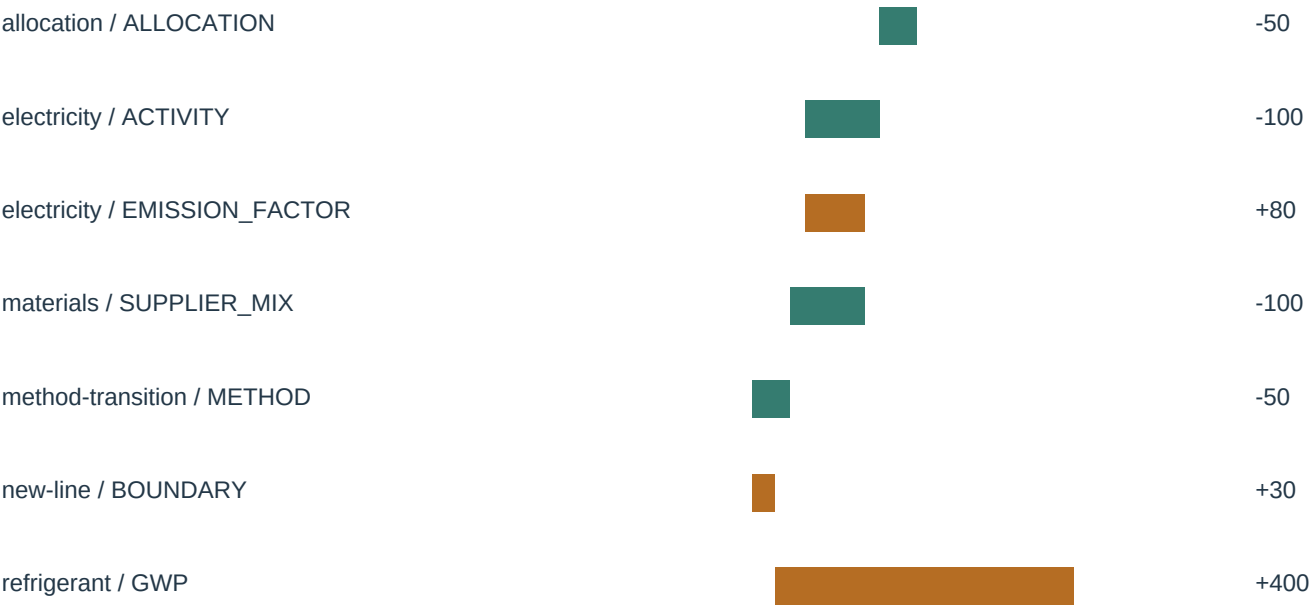
Declarations supply CORRECTION and MISSING_DATA_RESOLVED interpretations. The unassisted algorithm does not discover these business reasons.

INTERPRET / Zero assisted UNKNOWN reflects supplied assertions, not independently proven causes.

NEXT / Read the annual comparison using its declared decomposition convention.

08 / Reconcile the +210 change

Assisted waterfall: 3850 -> 4060 kgCO2e.



Electricity: $(800-1000) \times 0.5 = -100$ activity; $800 \times (0.6-0.5) = +80$ factor. Net -20. Activity-first, factor-second, allocation-last is the selected convention.

Other components: supplier mix -100, method -50, refrigerant GWP +400, boundary +30 and allocation -50. These include methodological changes, not only physical changes.

INTERPRET / Bars bridge cumulative values in report order. Components sum to +210; another convention can distribute interactions differently.

NEXT / Inspect UNKNOWN before accepting a narrative.

09 / Keep uncertainty visible

Compare unassisted and declaration-assisted outputs.

2025-v2 -> 2026-v1	Unassisted	Assisted
Delta	210.00	210.00
Attributed	-90.0	210.00
Signed UNKNOWN	300.00	0
Absolute UNKNOWN	500.00	0

Signed UNKNOWN can cancel: +400 and -100 yield +300 signed but 500 absolute exposure. Absolute exposure is not an extra emission total.

```
OUTPUT/declarations/2025-v2--2026-v1.json
OUTPUT/sources/scenarios/2025-v2--2026-v1/
OUTPUT/diffs/2025-v2--2026-v1/assisted.json
```

Read each rationale and its evidence path#sha256 reference. The runner checks the referenced scenario bytes. It does not authenticate a real-world cause.

INTERPRET / Never replace UNKNOWN with a plausible story. Assisted labels remain supplied interpretations.

NEXT / Preserve the complete run and verify its package.

10 / Verify the evidence package

Read the actual package verification result.

```
uv run --locked ghgag package verify
artifacts/worked-example/packages/2025-v1
```

```
{
  "valid": true,
  "record_count": 158,
  "manifest_sha256":
    "8ce82f3b4be8fa92027400f5f8744a2982ff0ebb029c20a295d7d52df7310eb6"
}
```

Location	What it preserves
packages/2025-v1/	JSON, graph, schema, ontology, manifest
sources/2025-v1/	15 synthetic source JSON documents
input-provenance.json	45 paths and SHA-256 digests
manifest.json	Full run hashes, except itself
commands.json	73 argv / exit / stdout / stderr receipts

Join command lines with a space. Sources remain outside the closed-profile crates. Preserve the whole tree for source documents, comparisons and receipts.

INTERPRET / Hashes detect changed bytes, not truth or authenticity. A substituted file plus substituted manifest can still agree.

NEXT / Use REPORT.md as an index and JSON for independent checks.

11 / Read the generated report

Actual REPORT.md excerpt, typeset here; not an app screenshot.

```
# ExampleCo-GHG-001 synthetic worked example

Generated by the installed CLI; selected sources only, not a complete corporate
inventory.
Invented factors, GWP bases and review decisions; no assurance opinion or causal
inference.
Integrity checks are not evidence authentication. No market-based fallback, offsets or
removals.

## Execution

Software: {'ghg-assurance-graph': '0.4.0a1', 'pydantic': '2.13.5', 'rdflib': '7.6.0',
'pyshacl': '0.31.0', 'rocrate': '0.15.1'}; Python 3.12.14.
Python socket connections, DNS and datagram sends blocked; not an OS sandbox. No
downloads or installation occur in this runner.
73 CLI receipts: [commands.json](commands.json). Paths are relative to OUTPUT.

## Partial totals and validation

| Snapshot | kg CO2e | Conforms (SHACL + raw rows) | Crate valid | Vault notes |
|---|---:|---|---|---:|
```

Annotation	How to read it
Profile / warning	Synthetic and completeness limits
Totals	Reconcile summary.json and 45 rows
Checks	Selected validation and package results
Later sections	Every formula, both diff modes, failures

The full report has 15 calculations per snapshot. `calculations/<snapshot>.json` preserves operands; `stages/<snapshot>/explain/` has every result lineage.

INTERPRET / A concise report is an evidence index, not a replacement. This guide does not print all 73 receipts or all 45 formulas.

NEXT / Inspect portable notes without assuming a desktop UI was tested.

12 / Inspect linked notes

Open OUTPUT/vaults/2025-v1/index.md in a Markdown reader.

```
uv run --locked ghgag export obsidian
  artifacts/exercise-vault
  --input artifacts/worked-example/generated/2025-v1
```

Portable export	Inspection route
Markdown	Start at index.md; follow [[note]] links
Turtle / JSON-LD	Graph exports and package artifacts
Canonical JSON	generated/<snapshot>/package.json

Join command lines with spaces. Optionally open the exported folder as an Obsidian vault, then select index. No plugins or cloud sync are required. Desktop UI behavior is not live-verified; no application screenshots are fabricated.

The runner verifies every generated wikilink target exists. This proves on-disk resolution, not rendering or interoperability in every editor.

INTERPRET / An evidence graph is not an assurance opinion. Share only authorized files; this example is fictional.

NEXT / Try a controlled input change in an isolated directory.

13 / Change one input safely

Predict first, then regenerate dependent evidence.

```
uv run --locked python scripts/modify_example.py
  --out artifacts/electricity-exercise
uv run --locked ghgag validate artifacts/electricity-exercise
```



Join the first two lines. The exercise changes 2025-v1 electricity from 1000 to 1100 kWh. Prediction: $+100 \times 0.5 = +50$ kgCO₂e; electricity 550; total 3870. It regenerates reported amount, canonical records and embedded evidence hashes, then checks Fraction arithmetic and validators.

This isolated teaching branch reuses the fixture namespace: never merge it with the original package. For export or packaging, use the new input directory and fresh destinations.

Do not edit derived results or hashes. benchmark generate restores the fixed baseline, not your exercise. Old crates and reports must not be reused for changed inputs.

INTERPRET / Expected: conforms=true and findings=[]. The script refuses existing output and leaves committed fixtures unchanged.

NEXT / Run troubleshooting checks before sharing.

14 / Troubleshoot transparently

Find the cheapest discriminating check.

Symptom	Check / safe recovery
uv not found	Install uv or use existing .venv/bin/uv
Wrong Python	Select 3.12 and sync the lock
Output refused	Choose a new or empty folder
Unexpected exit 1	Read rule, target and stderr receipt
Digest mismatch	Regenerate; never patch the manifest
UNKNOWN remains	Retain it; require explicit evidence
Wrong total	Check unit, share, precharacterization
Broken note link	Regenerate export from matching input

Do not suppress negative controls: two intentional exit-1 reports prove rejection paths execute. An unexpected failure is not successful reproduction.

Unsupported: physical-gas characterization, market-factor eligibility proof, complete Scope 3 screening, uncertainty quantification, source authentication and human assurance. No full ISO 14064-1 conformity or certification is claimed.

INTERPRET / Passing a bounded contract cannot establish facts that were never encoded or supplied.

NEXT / Finish with a second reviewer and preserve the source revision.

15 / Review, preserve, hand off

Use this checklist before sharing the evidence.

Check	Expected evidence
Revision + lock	Git SHA, uv.lock, Python version
Totals	3820 / 3850 / 4060 kgCO2e
Formulas	45 independent Fraction products
Receipts	73 actions; two expected exit-1 controls
Sources	45 matching source digests
Changes	+30 and +210; UNKNOWN disclosed
Packages	158 records per snapshot
Limits	Fictional / partial / no human assurance

Glossary: lineage = typed operand/evidence links; snapshot = one fixture version; CO2e = precharacterized equivalent emissions; UNKNOWN = unattributed change; declaration = supplied interpretation; SHA-256 = byte digest; RO-Crate = portable packaging.

Name choice: CompanyX is a real cyber-insurance business (KHIPU Networks partnership, 14 August 2025). Exact-label search on 29 September 2026 found no results for ExampleCo-GHG-001. This is a fictional identifier, not trademark or global-availability clearance.

Source: <https://www.khipu-networks.com/news/companyx-partnership/>

INTERPRET / Historical releases retain original names and hashes. Current regenerated fixtures use ExampleCo-GHG-001; arithmetic and schema are unchanged.

NEXT / See examples/worked_example/README.md for the full field dictionary and commands.